6. Scenario: You are a cashier at a grocery store and need to calculate the total cost of a customer's purchase, including applicable discounts and taxes. You have the item prices and quantities in separate lists, and the discount and tax rates are given as percentages. Your task is to calculate the total cost for the customer. Question: Use arithmetic operations to calculate the total cost of a customer's purchase, including discounts and taxes, given the item prices, quantities, discount rate, and tax rate?

AIM:

To calculate the total cost of a customer's purchase, including discount and tax, using Python arithmetic operations.

PROCEDURE:

1. Initialize item prices, quantities, discount rate, and tax rate.
2. Calculate the total price of all items.
3. Calculate the discount amount and subtract it from the total.
4. Calculate tax on the discounted amount.
5. Display the final cost.

PROGRAM CODE:

```
prices = [100, 200, 150]

quantities = [2, 1, 3]

discount_rate = 10

tax_rate = 5

subtotal = sum(p * q for p, q in zip(prices, quantities))

discount = subtotal * discount_rate / 100
```

after_discount = subtotal - discount

tax = after_discount * tax_rate / 100

total_cost = after_discount + tax

print("Subtotal:", subtotal)

print("Discount:", discount)

print("Tax:", tax)

print("Total Cost:", total_cost)

INPUT:

- Prices = [100, 200, 150]
- Quantities = [2, 1, 3]
- Discount = 10%
- Tax = 5%

OUTPUT:

```
    tax_rate = 5

    subtotal = sum(p * q for p, q in zip(prices, quantities))
    discount = subtotal * discount_rate / 100
    after_discount = subtotal - discount
    tax = after_discount * tax_rate / 100
    total_cost = after_discount + tax

    print("Subtotal:", subtotal)
    print("Discount:", discount)
    print("Tax:", tax)
    print("Total Cost:", total_cost)

...  Subtotal: 850
    Discount: 85.0
    Tax: 38.25
    Total Cost: 803.25
```

RESULT:

Thus, the final purchase cost after applying the discount and tax is 803.25.